

Patient Profile

Mr. Saubhik Bhaumik, 27-year-old male

Summary of Results

The liver function profile is within normal limits based on the provided parameters.

The predictive model (ML Result) assesses this profile as **Normal**, indicating no evident biochemical signs of liver dysfunction in the available data.

Detailed Test Explanations

SERUM BILIRUBIN (TOTAL)

- Patient result: 0.9 mg/dl
- Reference range: 0.2–1.2 mg/dl

Bilirubin is a breakdown product of red blood cells processed by the liver. Total bilirubin includes both direct (conjugated) and indirect (unconjugated) fractions.

- Elevated levels may indicate liver cell injury (hepatitis), bile duct obstruction, hemolysis, or hereditary bilirubin metabolism disorders.
- Low or normal values are generally not clinically concerning.

Your value of 0.9 mg/dl is within the normal range, suggesting no evidence of clinically significant jaundice or major bilirubin handling abnormality.

SERUM BILIRUBIN (DIRECT)

- Patient result: 0.2 mg/dl
- Reference range: 0–0.3 mg/dl

Direct (conjugated) bilirubin is bilirubin that has been processed by the liver and is excreted into bile.

- Elevated direct bilirubin can be seen in cholestasis (bile flow obstruction), hepatocellular disease, or certain hereditary disorders.

Your result of 0.2 mg/dl is within the normal range, suggesting no indication of significant cholestasis or impaired conjugation.

SERUM BILIRUBIN (INDIRECT)

- Patient result: 0.70 mg/dl
- Reference range: 0.2–1.0 mg/dl

Indirect (unconjugated) bilirubin reflects bilirubin before it is processed by the liver.

- Elevated levels may occur in hemolysis, fasting, or inherited conditions such as Gilbert syndrome, or in severe liver dysfunction affecting uptake/conjugation.

Your result of 0.70 mg/dl is within the normal range, indicating no evident problem with bilirubin production or pre-hepatic processing.

SGPT (ALT)

- Patient result: 36 U/L
- Reference range: 13–40 U/L

Alanine aminotransferase (ALT) is an enzyme primarily located in liver cells.

- Elevated ALT is a sensitive indicator of hepatocellular injury, as seen in viral hepatitis, fatty liver disease, medication-induced liver damage, and other liver disorders.
- Mild variations within the reference range are typically not clinically significant.

Your ALT of 36 U/L lies within normal limits and does not suggest active liver cell injury.

SGOT (AST)

- Patient result: 32 U/L
- Reference range: 0–37 U/L

Aspartate aminotransferase (AST) is found in liver, heart, muscle, and other tissues.

- Elevated AST can indicate liver injury, but may also increase in muscle damage or cardiac conditions.
- It is usually interpreted together with ALT and their ratio.

Your AST of 32 U/L is within the normal range and, together with a normal ALT, does not suggest ongoing significant hepatocellular damage.

SERUM ALKALINE PHOSPHATASE

- Patient result: Not reported in the provided data
- Reference range: Not provided

Alkaline phosphatase (ALP) originates from bile ducts, bone, and other tissues.

- Elevated ALP may indicate bile duct obstruction, cholestasis, or bone disease.
- Low values are less common and usually of lesser clinical significance.

Interpretation for this parameter cannot be made from the current dataset as no value has been reported.

SERUM PROTEIN (TOTAL PROTEIN)

- Patient result: 7.2 g/dl
- Reference range: 6.4–8.3 g/dl

Total serum protein comprises albumin and globulins.

- Low levels can be seen in malnutrition, chronic liver disease (reduced synthesis), renal protein loss, or malabsorption.
- High levels may suggest chronic inflammation, infection, or certain blood disorders.

Your value of 7.2 g/dl is within the normal range, suggesting adequate overall protein status and preserved synthetic function of the liver.

SERUM ALBUMIN

- Patient result: 4.7 g/dl
- Reference range: 3.5–5.2 g/dl

Albumin is the main protein made by the liver and is a key marker of liver synthetic capacity and nutritional status.

- Low albumin may indicate chronic liver disease (cirrhosis), nephrotic syndrome, protein-losing states, or malnutrition.
- Normal levels generally reflect preserved liver synthetic function.

Your albumin level of 4.7 g/dl is comfortably within the normal range, supporting good liver synthetic function and nutritional status.

GLOBULIN

- Patient result: 2.50 g/dl
- Reference range: 1.8–3.6 g/dl

Globulins include various proteins such as immunoglobulins and acute-phase reactants.

- Elevated globulins can occur in chronic inflammation, autoimmune diseases, chronic liver disease, or certain hematologic conditions.
- Low globulin levels may be associated with immune deficiencies or protein loss.

Your globulin level of 2.50 g/dl is within the normal range, not suggestive of chronic inflammatory or immune-mediated liver disease.

A/G RATIO (Albumin/Globulin Ratio)

- Patient result: 1.88
- Reference range: 1.1–2.4

The A/G ratio compares albumin to globulin, helping to assess the pattern of protein distribution.

- A low A/G ratio may be seen in chronic liver disease, autoimmune or inflammatory disorders, or certain gammopathies.
- A high A/G ratio can be associated with low globulin states.

Your A/G ratio of 1.88 is normal, consistent with balanced albumin and globulin levels and no apparent chronic hepatic or inflammatory dysfunction in this respect.

Overall Interpretation

The available liver function test parameters, including bilirubin fractions, transaminases (ALT, AST), total protein, albumin, globulin, and A/G ratio, are all within their respective reference ranges. This profile is consistent with:

- No biochemical evidence of acute hepatocellular injury
- No indication of significant cholestasis or bile duct obstruction based on bilirubin fractions
- Preserved liver synthetic function as reflected by normal albumin and total protein
- No pattern suggestive of chronic inflammatory or immune-mediated liver disease in the protein profile

The only limitation is the absence of a reported alkaline phosphatase value, which slightly restricts the assessment of bile duct and cholestatic pathology; however, with normal bilirubin fractions and transaminases, significant occult cholestasis is unlikely.

The predictive model (ML Result) classifies this profile as **Normal**, which is in agreement with the laboratory values and supports the interpretation that there is no current biochemical indication of liver disease in the parameters provided. Clinical context, symptoms, medications, and risk factors should still be considered by your healthcare provider.

Recommendations

1. Clinical correlation

- Share these results with your physician, especially if you have any symptoms such as fatigue, abdominal pain, jaundice, dark urine, pale stools, itching, or unexplained weight loss.
- If asymptomatic and with no liver-related risk factors, these results are reassuring.

2. Lifestyle and liver health

- Maintain a balanced diet rich in fruits, vegetables, whole grains, lean proteins, and healthy fats.
- Limit excessive intake of saturated fats, ultra-processed foods, and added sugars to reduce the risk of fatty liver disease.
- Avoid smoking and limit alcohol consumption; if you drink, do so within recommended safe limits, or consider abstinence, especially if there are any liver risk factors.

3. Medication and hepatotoxic substances

- Use over-the-counter painkillers (especially paracetamol/acetaminophen and certain NSAIDs) only as directed and avoid self-medication at high doses or for prolonged periods.
- Inform your doctor of any herbal supplements, bodybuilding or performance-enhancing products, or traditional remedies you use, as some can be hepatotoxic.

4. Weight, exercise, and metabolic health

- Aim for a healthy body weight and waist circumference through regular physical activity and diet.
- Engage in at least 150 minutes per week of moderate-intensity aerobic exercise plus muscle-strengthening activities on 2 or more days per week, if medically appropriate.
- If you have diabetes, high blood pressure, or high cholesterol, ensure they are well controlled, as these conditions increase the risk of non-alcoholic fatty liver disease.

5. Follow-up testing

- With a normal ML Result and normal parameters, routine re-testing is typically recommended only as part of periodic health check-ups, or sooner if:
 - Liver-related symptoms develop.
 - You start new long-term medications known to affect the liver.
 - You have risk factors such as heavy alcohol use, viral hepatitis exposure, or metabolic syndrome.
- If there is a specific clinical concern (e.g., persistent symptoms, known liver disease, or strong family history), your physician may consider completing the profile with alkaline phosphatase, GGT, or imaging (such as ultrasound) as appropriate.

Disclaimer

This report is an informational interpretation of the provided laboratory data and predictive model output and does not replace an in-person clinical evaluation. Diagnosis and treatment decisions must be made by a licensed healthcare professional who can consider your full medical history, examination findings, and any additional investigations.

